

Quantum Computing Research Paper

Advanced Studies in Quantum Mechanics

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Abstract

This research investigates the quantum properties of photonic crystals and their applications in quantum computing. Our study demonstrates significant improvements in quantum coherence time and error correction capabilities.

1. Introduction

Quantum computing represents a paradigm shift in computational theory and practice. The fundamental principle relies on quantum superposition and entanglement to perform calculations exponentially faster than classical computers for certain problem classes.

2. Theoretical Framework

The quantum state is described by:

$$|\psi\rangle = \alpha|0\rangle + \beta|1\rangle$$

where $|\alpha|^2 + |\beta|^2 = 1$

The Hamiltonian energy:

$$E = \hbar\omega(n + 1/2)$$

3. Experimental Results

Our experiments demonstrate a coherence time of 150 microseconds, representing a 40% improvement over previous results. The quantum gate fidelity achieved was 99.7%, exceeding the fault-tolerant threshold required for practical quantum computing.

Table 1: Experimental Measurements

Parameter	Value	Unit
Coherence Time	150	·s
Gate Fidelity	99.7	%
Qubit Count	127	qubits
Error Rate	0.003	per gate

4. Discussion

These results indicate significant progress in quantum computing hardware. The improved coherence time enables more complex quantum algorithms to be executed before decoherence occurs.

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5. Conclusion

In conclusion, our research demonstrates substantial improvements in quantum computing hardware performance. The 40% increase in coherence time and 99.7% gate fidelity represent significant milestones toward practical quantum computing applications.

Future work will focus on scaling up the qubit count while maintaining these performance metrics, and developing error correction protocols optimized for our hardware architecture.

References

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